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To the Editor-in-Chief,
ACM Transactions on Mathematical Software

Re: Submission of the manuscript "Tuning Internal Parameters of Algebraic Multigrid: A Cross-Library Study of Achievable Speedup and Cheap-Feature Prediction"

Dear Editor,

Please consider the enclosed manuscript for publication in ACM Transactions on Mathematical Software as a research article.

Algebraic multigrid (AMG) is the workhorse preconditioner embedded in every major sparse-solver framework, yet its performance depends strongly on internal parameters — coarsening scheme, smoother, and strength-of-connection threshold — that most users leave at library defaults. Prior algorithm-selection work almost exclusively chooses *among* solvers or preconditioners, treating an entire AMG method as a single label. Our manuscript instead opens that label and studies the interior parameter space empirically: we sweep 88 configurations of AMGCL over 150 (later 300) SuiteSparse matrices — 13,200 timed solves — and ask how much speed is lost by not tuning, and whether a good configuration can be predicted from features that are cheap relative to the solve itself.

The main findings are:

- **Tuning matters and requires per-matrix selection.** The oracle speedup over the default has median 2.4× (upper quartile 6.4×, maximum 292×); the best configuration varies across 14 distinct coarsening/smoother labels, and on 31 matrices the default fails outright while some configuration succeeds.
- **A predictor works from near-free features, evaluated honestly.** Under a strict leave-one-group-out protocol — no sibling matrices leaking across the split, a check the literature commonly omits — the predictor picks a configuration that solves 83–89% of held-out matrices and captures a median 43–49% of the achievable speedup. Near-free structural features suffice; spectral features costing 100–1000× more add nothing, a feature-cost accounting that prior work seldom performs.
- **A self-critical cross-library validation.** Repeating the study on hypre BoomerAMG shows that tunability replicates qualitatively but its magnitude is smaller (median 1.6×) and does not transfer between libraries (rank correlation 0.24); it is largely governed by default quality — the dramatic AMGCL speedups occur precisely where hypre's well-engineered default already wins. Intrinsic solvability, by contrast, is library-independent (hard-matrix Jaccard 0.76), a split that a predictor-transfer experiment confirms in both directions.

We believe the work fits TOMS well on both scope and values. It is an empirical study of two widely-used pieces of mathematical software (AMGCL and hypre), it foregrounds honest methodology (feature-cost accounting and group-wise evaluation), and it is fully reproducible. We release the complete artifact — the 18,600-solve dataset, the feature extractor, the predictor, and all figure-generation scripts — publicly at `https://github.com/mirryou-maker/amg-tuning-crosslib`. A single command, `python reproduce.py`, regenerates every figure and headline number from the released data, and a companion tool, `tools/recommend.py`, applies the predictor to a user's own matrix. We would welcome consideration under the Replicated Computational Results process.

We confirm that this manuscript is original, has not been published previously, and is not under consideration elsewhere. In accordance with the ACM Policy on Authorship, the manuscript includes a Generative AI disclosure describing the use of an AI coding assistant (Claude Code, Anthropic) for implementation, experiment orchestration, and drafting; the author verified all experimental design, numerical results, and claims against the released code and data, and takes full responsibility for the content. The work was supported by the National Research Foundation of Korea (Nos. RS-2025-25463492 and RS-2026-25472340) and the DGIST R&D Program (26-SR-01).

Suggested reviewers. We suggest the following experts in algebraic multigrid, solver selection, and numerical software (contact details available on request); we have no conflict of interest with them:

- **Luke N. Olson**, University of Illinois Urbana-Champaign — algebraic multigrid and the PyAMG software.
- **Jed Brown**, University of Colorado Boulder — multigrid, PETSc, and numerical-software performance.
- **Kanika Sood**, California State University, Fullerton — machine learning for iterative-solver selection (the closest prior line of work).
- **Scott P. MacLachlan**, Memorial University of Newfoundland — multigrid methods and their analysis.
- **Sivasankaran Rajamanickam**, Sandia National Laboratories — sparse linear algebra, performance, and learning-based methods for solvers.

Thank you for considering our submission.

Sincerely,

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